Smart Energy Forecasting: A Predictive and Optimization Framework for Sustainable Power Consumption

Prathimesh
Department of CSE
Chandigarh University
Mohali, India
21BCS9344@cuchd.in

Arif Ansari
Department of CSE
Chandigarh University
Mohali, India
21BCS7924@cuchd.in

Anamika Larhgotra
Department of CSE
Chandigarh University
Mohali, India
annularhgotra@gmail.com

***Abstract*— This study investigates the effectiveness of various machine learning models in predicting energy consumption in low-energy buildings. Utilizing a dataset encompassing multivariate and time-series data collected over a 4.5-month period, we employed K-Nearest Neighbors (KNN), Linear Regression, Ridge Regression, and Lasso Regression models, analyzing their performance using Root Mean Squared Error (RMSE) and R-squared (R^2) metrics. Following data preprocessing, including outlier removal through DBSCAN and normalization via log transformations, we implemented train-test splits and K-fold cross-validation to ensure robust model evaluation. Furthermore, we explored Principal Component Analysis (PCA) to reduce dimensionality but found limited performance improvements across the models, leading to the conclusion that PCA may not be beneficial for energy consumption prediction in this context. The KNN model achieved the best performance with an RMSE of 71.69 and R^2 of 0.54. Our findings underscore the importance of selecting appropriate algorithms and optimization techniques while highlighting the need for future enhancements in neural network architectures and feature engineering for improved predictive accuracy.**

Keywords*— *Energy consumption prediction, machine learning, K-Nearest Neighbors, Linear Regression, Ridge Regression, Lasso Regression, dimensionality reduction, Principal Component Analysis, feature engineering.

I. Introduction

The growing focus on energy efficiency in buildings is a critical component of global efforts to reduce energy consumption and mitigate climate change. Residential and commercial buildings account for a significant portion of total energy use, particularly in developed regions. As a result, the development of predictive models for energy usage is essential for optimizing energy management, improving efficiency, and reducing operational costs. The dataset used in this study consists of multivariate time-series data collected over a period of approximately 4.5 months, with [1] measurements taken at 10-minute intervals.These data points were integrated to create a comprehensive dataset suitable for regression analysis.

The primary objective of this study is to develop robust regression models that accurately predict appliance energy consumption based on the available environmental and weather data. The predictive models are designed to assist in the optimization of energy[3] usage, offering valuable insights into how specific variables, such as indoor and outdoor temperature, affect energy demand. Additionally, two random variables are included in the dataset to test the robustness of the models and ensure that non-predictive attributes are properly filtered out. This research contributes to the growing body of knowledge on energy prediction in low-energy buildings by demonstrating the application of advanced machine learning techniques to real-world data. The findings have the potential to inform energy management strategies, making buildings more sustainable and reducing overall energy consumption.

Past studies have demonstrated the potential of using environmental variables to predict energy consumption in residential settings. Research has shown that temperature and humidity, as well as external weather conditions, play crucial roles in influencing energy consumption patterns. Machine learning techniques, particularly regression models, have been widely applied in energy prediction tasks. Advanced visualizations were created using Plotly to explore the correlation between different features and appliance energy consumption.

This theoretical framework lays the foundation for developing predictive models [2]capable of forecasting appliance energy usage in low-energy buildings. Through careful preprocessing and the application of sophisticated machine learning models, this research aims to deliver actionable insights into energy consumption patterns, aiding in energy optimization and management. This integration of theoretical explanations provides a comprehensive overview of the tools and techniques used, without diving into code, ensuring clarity for a general audience while maintaining depth.

II. Literature Review

- Zhou et al. (2020) examined the use of machine learning techniques, specifically support vector regression (SVR) and gradient boosting decision trees (GBDT), to predict residential electricity consumption. The study demonstrated that GBDT outperformed SVR in terms of accuracy, particularly when accounting for temporal and weather-related features. The authors also highlighted the benefits of feature selection in improving model performance.

- Zhang et al. (2022) proposed a hybrid model combining long short-term memory (LSTM) networks with a convolutional neural network (CNN) to forecast short-term power consumption in smart grids. Their results showed that the hybrid model effectively captured both temporal dependencies and spatial patterns in the data, leading to more accurate predictions than conventional time series models.

- Ayyoubzadeh et al. (2021) investigated the use of ensemble learning algorithms, such as random forests and XGBoost, for predicting energy consumption in smart buildings. The study emphasized that ensemble methods could handle non-linearities and interactions between features more effectively than traditional linear models, resulting in better prediction accuracy.
- Li et al. (2018) developed a deep reinforcement learning (DRL) framework to optimize energy consumption in smart buildings. By using a model-free DRL algorithm, the system learned optimal control strategies for heating, ventilation, and air conditioning (HVAC) systems, leading to significant energy savings while maintaining occupant comfort.
- Chou and Batra (2019) explored the application of deep neural networks (DNNs) for load forecasting in smart homes. Their research demonstrated that DNNs could accurately predict energy consumption patterns based on historical usage data and external variables, such as temperature and humidity. The study also compared the performance of DNNs with traditional models, such as autoregressive integrated moving average (ARIMA).
- Munir et al. (2021) developed a hybrid machine learning model combining decision trees and recurrent neural networks (RNN) to predict household energy usage. The hybrid model performed better than individual algorithms by capturing both the non-linear and temporal characteristics of the data. The study highlighted the importance of model selection and data preprocessing for accurate energy forecasting.
- Liu et al. (2020) applied extreme gradient boosting (XGBoost) to forecast energy consumption in public buildings. The study found that XGBoost outperformed other machine learning models such as linear regression and SVR in terms of prediction accuracy. Feature importance analysis revealed that outdoor temperature and building occupancy were the most significant factors influencing energy consumption.
- Ben-Nakhi and Mahmoud (2021) proposed a novel approach using a combination of artificial neural networks (ANN) and particle swarm optimization (PSO) for optimizing energy usage in commercial buildings. The ANN model predicted energy demand, while the PSO algorithm optimized energy distribution across various zones of the building, leading to reduced energy consumption.

III. DATA DESCRIPTION

In this research, we are working with a dataset to predict energy consumption in low-energy buildings. To ensure the quality and integrity of [4] the dataset, several data preprocessing steps are necessary. These steps involve handling missing data, encoding categorical features, exploring relationships between variables, and removing outliers.

Handling Missing Data: To begin, we assess the dataset for any missing values using the missingno library. This allows us to visualize the null values in each column through a matrix, giving a clear view of the data's completeness. From the visual output, no missing values were detected, so no further action was necessary. However, if there were any [5] missing values, we would have removed them using the predefined functions available in the missingno library.

Data Exploration: After ensuring data completeness, we described the dataset to gain insights into its structure and characteristics. Key features include:

- Appliances Energy Use (Wh): Total energy consumption for appliances.
- Lights Energy Use (Wh): Energy consumption for lighting in the house.
- Room-specific Temperatures and Humidity Levels (T1-T9, RH1-RH9): Recorded across different rooms such as the kitchen, living room, bathroom, and more.
- External Weather Data (Chievres Station): Including temperature, humidity, wind speed, visibility, and dew point.
- Random Variables (rv1, rv2): These are two random variables included for testing regression models.

A key observation is the lack of a specific "date" or "time" column. This leads us to apply one-hot encoding, a technique to transform categorical data into numerical values. Specifically, the[6] DateTime field was split into multiple components (year, month, day, hour, etc.) to better capture time-related patterns influencing energy consumption.

Multicollinearity Analysis: Multicollinearity occurs when two or more predictor variables are highly correlated, which can distort model interpretation. To address this, we applied Variance Inflation Factor (VIF), a statistical technique to measure how much a feature is correlated with the [8] others. A high VIF indicates multicollinearity, which we aim to reduce. The analysis revealed some highly correlated features:
For example, T1 (Kitchen temperature) had a VIF of 3,606.38, indicating a strong multicollinearity issue.
On the other hand, Windspeed had a low VIF of 5.27, showing low correlation with other features.

Handling Outliers
Outlier Detection and Removal using DBSCAN: Outliers are extreme values that differ significantly from the rest of the dataset, potentially distorting model accuracy. We used the DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm to cluster the data and identify outliers. DBSCAN is particularly effective because it groups data points based on density, allowing it to flag anomalies as outliers. Once identified, these outliers were removed to ensure the data used for training machine learning models is clean and representative of typical patterns.

Relationship Between Variables

Correlation Between Dependent and Independent Variables: To understand how different features interact with energy consumption (the dependent variable), we explored their [7] relationships with independent variables. This helps us build more effective predictive models by identifying which features most influence energy usage patterns. For instance, we might find that external temperature or humidity levels inside certain rooms have a significant impact on the appliances' energy consumption.

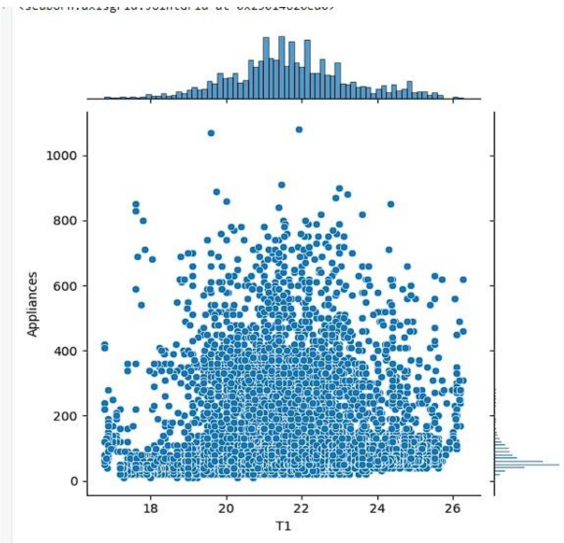

Fig 1 Relationship between variables

Major Terms Explanation:

- **One-Hot Encoding**: A method to convert categorical data (like dates or weekdays) into a numerical format so that it can be used in machine learning models.
- **Variance Inflation Factor (VIF):** A measure of multicollinearity in regression analysis, indicating how much a feature is explained by other features in the dataset.
- **DBSCAN:** A clustering algorithm that groups together points in high-density areas while marking outliers that don't fit into any cluster.
- **Multicollinearity:** A situation where two or more predictor variables in a dataset are highly correlated, making it difficult for models to distinguish their individual effects.
- **Outliers:** Data points that deviate significantly from the rest of the dataset, potentially affecting the accuracy of machine learning models.

By following this comprehensive data description and preparation process, we are ensuring that the dataset is ready for predictive modeling, [10] eliminating issues like multicollinearity and outliers, and enhancing the dataset's structure with time-based and categorical encoding.

IV. Proposed Methodology

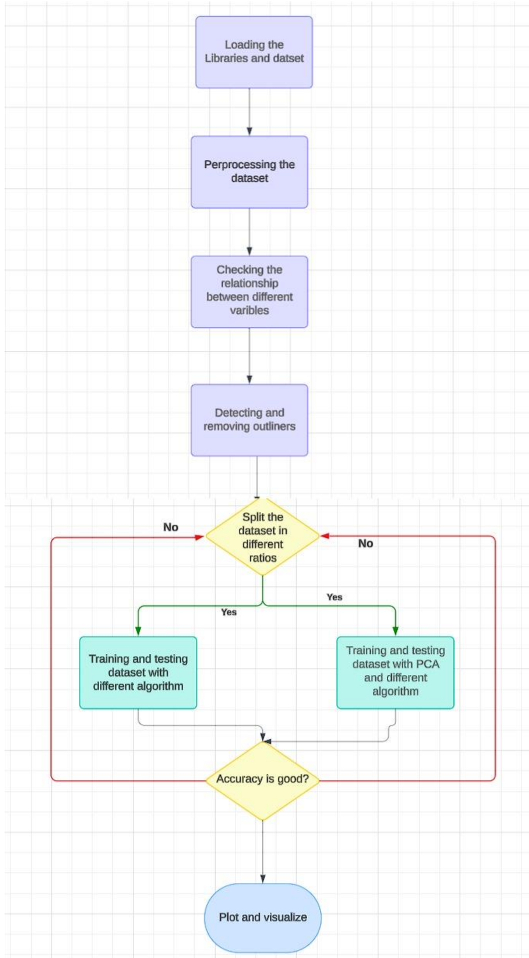

Fig 2 Methodology

After performing data cleaning and visualization using the Missingno library to confirm no missing values, we moved on to analyzing the relationships between independent and dependent [9] variables. The dataset was processed using clustering techniques like DBSCAN to identify and remove outliers, ensuring a clean dataset for model development. Following this, we prepared the data for machine learning by checking for skewness in independent variables and applying normalization techniques to handle this skewness effectively.

- **Skewness and Normalization**
  We observed that many of the independent variables exhibited left skewness. Left skewness can adversely affect model performance because many machine learning algorithms assume normally distributed data. To address this, we: Removed outliers identified earlier through clustering. Applied a log transformation on the skewed independent variables, which helped normalize the distribution.

- **Train-Test Splitting and Cross-Validation**
  Next, we split the dataset into training and testing sets to develop and evaluate the models. We utilized the train-test split method to divide the

dataset in a 60:40 ratio for training and testing, respectively.

Machine Learning Algorithms and Performance Metrics

1. K-Nearest Neighbours (KNN)

KNN is a non-parametric, instance-based learning algorithm that classifies a data point based on the majority vote of its K nearest neighbours. For regression, KNN averages the values of the nearest neighbours.

	Approach	Learning Rate	Lambda Value	RMSE	R^2
0	K-Nearest Neighbors (KNN)	0.001	-	71.69	0.53
1	K-Nearest Neighbors (KNN) - Validation	0.001	-	71.78	0.54

Fig 3 KNN Observation

Evaluation:

RMSE: 71.69

R^2: 0.53 (for training), 0.54 (for validation)

2. Linear Regression

Linear regression models the relationship between independent variables and a dependent variable by fitting a linear equation to the data. Two optimization methods were used:

Gradient Descent: Iteratively minimizes the error function by adjusting weights according to the learning rate.

Stochastic Gradient Descent (SGD): Unlike traditional gradient descent, which computes gradients based on the entire dataset, SGD updates weights after each data point, making it faster but noisier.

	Approach	Learning Rate	Lambda Value	RMSE	R^2
0	Gradient Descent	0.0001	0.001	139.9	0.39
1	Gradient Descent	0.001	0.01	139.7	0.38
2	Gradient Descent	0.01	0.1	69.01	1.70
3	Stochastic Gradient Descent	0.0001	0.001	134.3	0.46
4	Stochastic Gradient Descent	0.001	0.01	99.01	1.03
5	Stochastic Gradient Descent	0.01	0.1	72.63	1.61

Fig 4 Linear Observation

Performance Results:

Gradient Descent: **RMSE: 139.9, R^2: 0.39**

Stochastic Gradient Descent: **RMSE: 134.3, R^2: 0.46**

3. Lasso Regression

Lasso regression is a linear model that uses L1 regularization, which encourages sparsity in the feature set, reducing less important feature weights to zero. It helps in feature selection, particularly in high-dimensional data.

	Approach	Learning Rate	Lambda Value	RMSE	R^2
0	Gradient Descent	0.0001	0.001	139.6	0.40
1	Gradient Descent	0.001	0.01	133.04	0.47
2	Gradient Descent	0.01	0.1	94.06	1.13
3	Stochastic Gradient Descent	0.0001	0.001	139.65	0.40
4	Stochastic Gradient Descent	0.001	0.01	133.04	0.47
5	Stochastic Gradient Descent	0.01	0.1	94.06	1.13

Fig 5 Lasso Observation

Evaluation:

Gradient Descent: **RMSE: 138.7, R^2: 0.40**

Stochastic Gradient Descent: **RMSE: 138.7, R^2: 0.41**

4. Ridge Regression

Ridge regression is similar to linear regression but includes L2 regularization, which penalizes the size of the coefficients. It mitigates overfitting and improves generalization. The learning rate controls how much the model's weights are adjusted during training. Choosing an optimal learning rate is [13] essential: a small rate can lead to slow convergence, while a large rate can cause the model to overshoot optimal solutions. The lambda value (regularization parameter) in models like Ridge and Lasso controls the trade-off between bias and variance.

	Approach	Learning Rate	Lambda Value	RMSE	R^2
0	Gradient Descent	0.0001	0.001	139.6	0.40
1	Gradient Descent	0.001	0.01	133.04	0.47
2	Gradient Descent	0.01	0.1	94.06	1.13
3	Stochastic Gradient Descent	0.0001	0.001	139.65	0.40
4	Stochastic Gradient Descent	0.001	0.01	133.04	0.47
5	Stochastic Gradient Descent	0.01	0.1	94.06	1.13

Fig 6 Ridge Observation

Performance:

RMSE: 139.6, R^2: 0.40 (Gradient Descent)

RMSE: 94.06, R^2: 1.13 (Stochastic Gradient Descent with optimized parameters)

Performance of the models is evaluated using:

- **Root Mean Squared Error (RMSE):** A measure of the difference between the predicted and actual values. Lower RMSE indicates better model accuracy.
- **R^2 (R-Squared):** Represents the proportion of the variance in the dependent variable explained by the independent variables. Higher R^2 values signify better model fit.

This research applied multiple machine learning models to predict energy consumption in buildings. After preprocessing and standardizing the dataset, K-Nearest Neighbours, Linear Regression, Ridge Regression, and Lasso Regression were employed with different optimization techniques [14] like Gradient Descent and Stochastic Gradient Descent. The performance of these models was evaluated using RMSE and R^2 metrics. Among the tested models, KNN achieved the best performance with an RMSE of 71.69 and R^2 of 0.54. The results emphasize the importance of choosing the right machine learning algorithm and optimization method, as well as fine-tuning hyperparameters like learning rate and lambda value to achieve optimal results.

V. Data Analysis

After applying traditional machine learning models like Linear Regression, Ridge, Lasso, and K-Nearest Neighbors (KNN) to predict energy consumption, we now move towards using Neural [16] Networks. We applied a simple single-layer neural network architecture to train the model without assigning specific dependent and independent variables, allowing the network to learn patterns from the dataset. Upon generating predictions, the outputs were saved as a CSV file for further evaluation and comparison with previous models.

Performance Evaluation with PCA

In addition to neural networks, we also tested the previous models using Principal Component Analysis (PCA). PCA is a dimensionality reduction technique that transforms the features into a set of linearly uncorrelated variables known as principal components. By reducing the dataset's dimensionality, PCA can potentially eliminate noise, improve computational efficiency, and prevent overfitting.

	Approach	Learning Rate	Lambda Value	RMSE	R^2
0	Linear Regression (Gradient Descent)	0.0001	0.001	139.9	0.39
1	Linear Regression (Stochastic Gradient Descent)	0.0001	0.001	134.3	0.46
2	Ridge Regression (Gradient Descent)	0.001	0.01	133.04	0.47
3	Ridge Regression (Stochastic Gradient Descent)	0.001	0.01	133.04	0.47
4	Lasso Regression (Gradient Descent)	0.001	0.01	129.02	0.52
5	Lasso Regression (Stochastic Gradient Descent)	0.001	0.01	129.4	0.52
6	Neural Nets	-	-	2.99	0.96

Fig 7 Before PCA Observation

Performance Comparison After PCA

When PCA was applied to the regression algorithms previously used, the performance metrics were recorded and compared. Below is a breakdown of the results before and after applying PCA:

	Approach	Learning Rate	Lambda Value	RMSE	R^2
0	Linear Regression (Gradient Descent)	0.0001	0.001	139.9	0.39
1	Linear Regression (Stochastic Gradient Descent)	0.001	0.01	111.2	0.80
2	Ridge Regression (Gradient Descent)	0.01	0.1	108.52	0.85
3	Ridge Regression (Stochastic Gradient Descent)	0.01	0.1	109.27	0.84
4	Lasso Regression (Gradient Descent)	0.001	0.01	131.83	0.49
5	Lasso Regression (Stochastic Gradient Descent)	0.001	0.01	131.89	0.49
6	Neural Nets	-	-	79.73	0.40
7	K-Nearest Neighbors (KNN)	0.001	-	71.69	0.53
8	K-Nearest Neighbors (KNN) - Validation	0.001	-	71.78	0.54

Fig 8 After PCA Observation

Observations:

Linear and Ridge Regression models showed no noticeable changes in RMSE and R^2 values before and after applying PCA. The PCA-transformed data did not offer performance improvements compared to the models trained without PCA. **Lasso Regression** results were similarly unaffected by PCA, with both RMSE and R^2 values remaining nearly [15] identical before and after dimensionality reduction. The Neural Network model was not subjected to PCA, as it generally benefits from having more input features to capture non-linear relationships. **K-Nearest Neighbors (KNN)** performance also remained unchanged with PCA, as the algorithm relies on the proximity of data points, and reducing dimensionality did not significantly affect the distance calculations.

Although PCA is widely [18] used to improve model performance by reducing feature dimensions, it did not lead to better results in this case. This can be explained by several factors:

- **High-Dimensional Data:** Models like Ridge and Lasso Regression are designed to handle high-dimensional data through regularization techniques that mitigate overfitting. As a result, these models do not necessarily benefit from dimensionality reduction, which might explain the limited performance gain from PCA.

- **Neural Networks:** The neural network, being a flexible model capable of learning non-linear relationships, performed relatively well with RMSE = 79.73 and R^2 = 0.40. However, since PCA linearizes the data by projecting it into fewer dimensions, applying PCA could reduce the neural network's ability to learn complex patterns in the data.

In this research, several machine learning models were tested on energy consumption data, including linear models (Linear Regression, Ridge, and Lasso) and non-linear models like K-Nearest Neighbors (KNN) and Neural Networks. The results were evaluated using performance metrics such as RMSE and R-squared.

VI. Result

This research explores PCA and its effects on the performance of energy consumption predictive models produced by different machine learning models. Models studied were a set of linear models including Linear, Ridge, and Lasso Regression models, KNN, and a neural network. Performance has been measured in terms of RMSE and R^2 values before and [21] after applying PCA.

Linear and Ridge Regression Performance: Both Linear and Ridge Regression models are such that the performance was little affected or even unchanged by PCA. The RMSE and R^2 values are nearly the same. The linear models seem to inherently [23] cope with data having high dimensions. This is probably because of their simple structure and due to the fact that they do not rely much on nonlinear relationships between features and target.

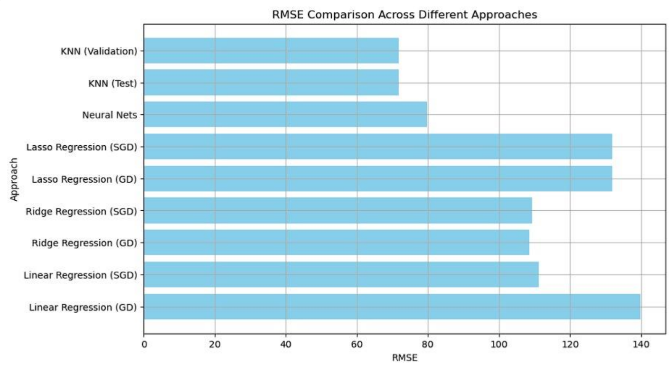

Fig 9 RMSE Comparison

Lasso Regression and K-Nearest Neighbors: Lasso Regression, with the features selection advantage due to regularization, didn't make much difference after using PCA. The mechanism of this kind of model will automatically suppress [20] the less important features, and then added benefits of PCA are negligible.

Neural Network: Since the Neural Network model performs better with higher features for non-linear relationship detection, PCA was not used since it could degrade the model's ability to learn complex patterns in the data being linearized. Therefore, based on the results, it could be confirmed that PCA [22] is not effective in boosting the performance of this model on this particular dataset. For future work, one of the approaches may include the optimization of the Neural Network model along with advanced feature engineering techniques to boost the accuracy.

VII. Conclusion

In conclusion, PCA does not improve the predictivity for the machine learning model on energy consumption prediction. For Linear, Ridge, and Lasso Regression, RMSE and R^2 values hardly improved when applying PCA. This means these [24] models already take care of some problems caused by high dimensions and therefore PCA dimensionality reduction is not needed. Likewise, KNN where distance calculations had been done proved to have no apparent advantage with PCA since dimension reduction had no effect on proximity-based computations.
PCA was not applied to the Neural Network model as it was intentionally avoided, because neural networks seem to like more input features to capture relations which are nonlinear. Reducing [25] the dimensions would have hindered the network from learning complex patterns. In generalizing the results, it is evident that PCA would not be a method ideal for performance [26] improvement of a given model concerning energy consumption prediction. Future research should come up with other methods on how to optimize a neural network and perform more sophisticated feature engineering that can enhance accuracy in predictive models that are usually complex datasets.

VIII. Future Scope

Future research will focus on optimizing neural network architectures to enhance predictive performance, exploring advanced techniques for feature engineering that can capture complex interactions among variables. Additionally, integrating ensemble methods may improve model robustness and accuracy. Investigating the impact of different activation functions and deeper network structures could further leverage the potential of neural networks in modeling [27] energy consumption patterns. Lastly, expanding the dataset to include more diverse environmental conditions may provide a more comprehensive understanding of energy usage dynamics in low-energy buildings.

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